

Bellabeat Case Study: Dashboard Visualization Report

FitBit Usage Analysis — Power BI Dashboard (5 pages, 13 visualizations)

Overview Page

Total Users (Card)

33 total FitBit users analyzed across activity, sleep, and heart rate data.

Sleep Data Available for 24 IDs (Card)

24 of 33 users (73%) log sleep data at all — the first sign of an engagement gap explored further in the Engagement pillar.

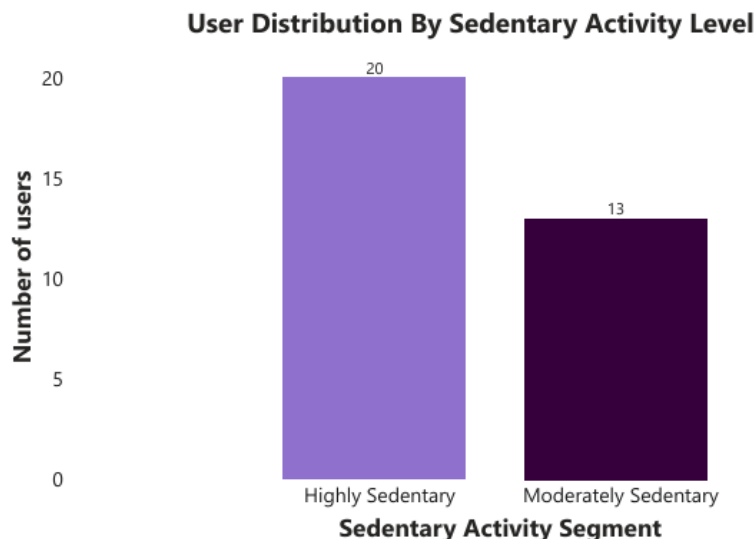
Steps – Calories Correlation (Card)

$r = 0.59$ (moderate positive). Steps alone don't fully explain calories burned, supporting Leaf's intensity tracking as a differentiator beyond simple step counts.

Resting HR – Activity Correlation (Card)

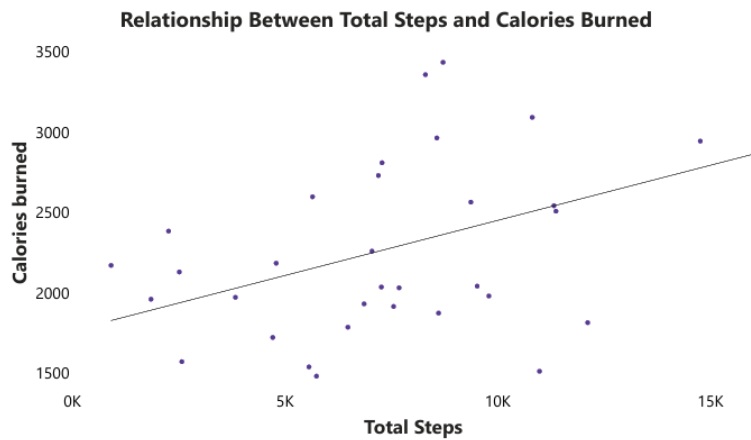
$r = -0.69$ (strong negative, $n = 14$). More active users show meaningfully lower resting heart rates — the dashboard's strongest and most differentiated finding.

Activity Pillar



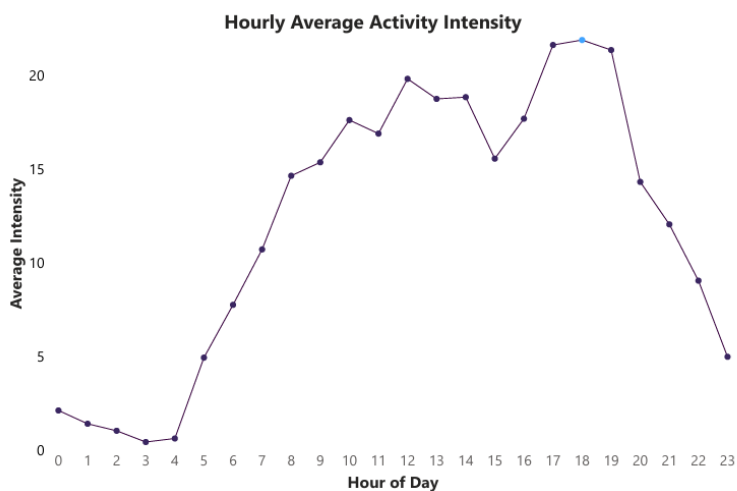
User Distribution by Sedentary Activity Level

Users split into two clear segments: 20 Highly Sedentary (~19 hrs/day) vs. 13 Moderately Sedentary (~12 hrs/day). Supports two distinct Leaf messaging strategies rather than one universal activity goal.



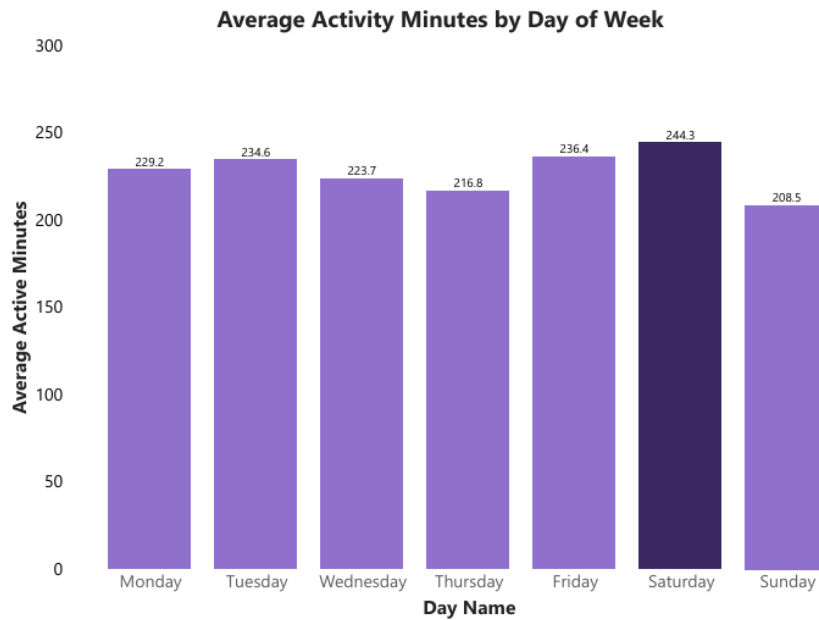
Relationship Between Total Steps and Calories Burned

Scatter plot with trend line, $r = 0.59$. A real but noisy upward relationship — calories burned depends on more than steps alone (body composition, non-step exercise).



Hourly Average Activity Intensity

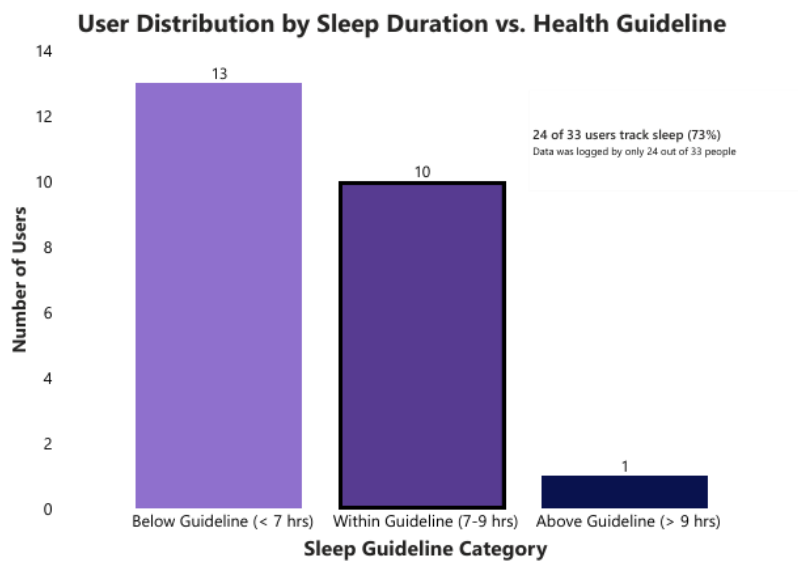
Bimodal daily rhythm: intensity peaks around 12–13 (lunchtime) and 18–19 (early evening), with a dip between and low overnight activity.



Average Activity Minutes by Day of Week

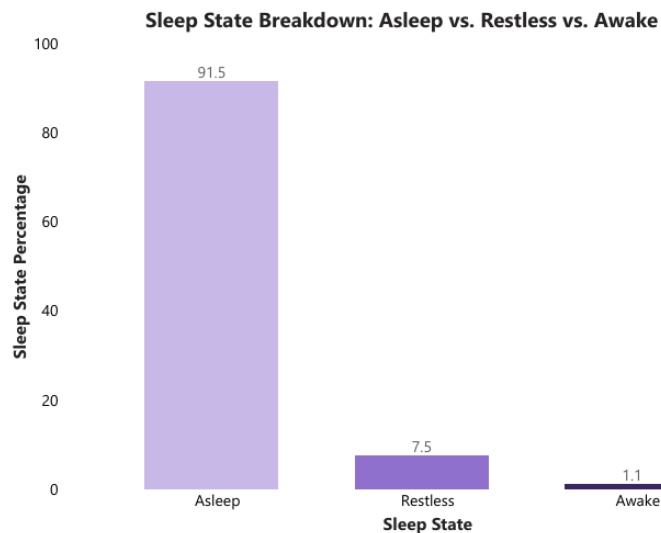
Tuesday highest (244.3 min), Sunday lowest (208.5 min). A modest but consistent weekly pattern, useful for timing weekend-specific engagement nudges.

Sleep Pillar



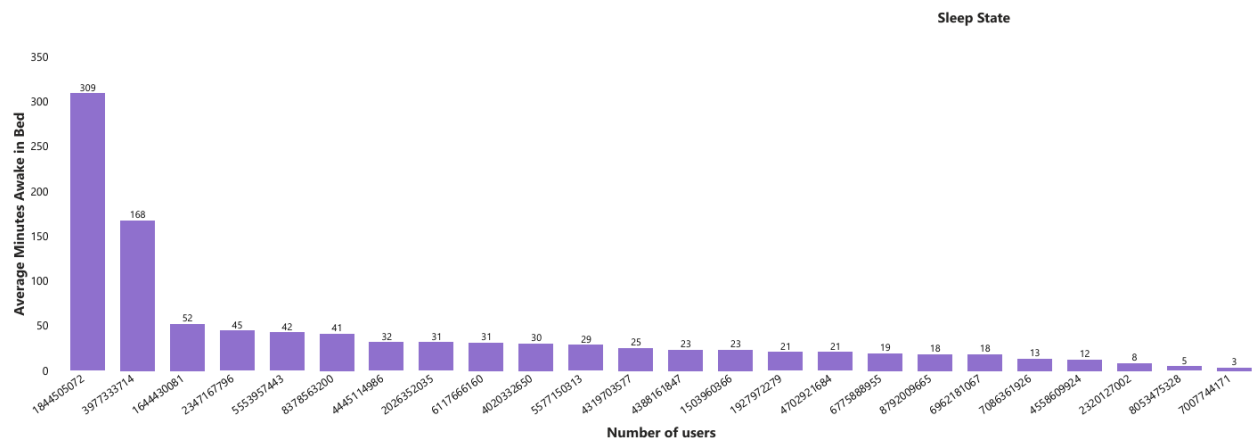
User Distribution by Sleep Duration vs. Health Guideline

Of 24 users with sleep data: 13 sleep below the 7-hour guideline, 10 within the 7–9 hour range, and just 1 above 9 hours. Most trackers are under-sleeping relative to recommendations.



Sleep State Breakdown: Asleep vs. Restless vs. Awake

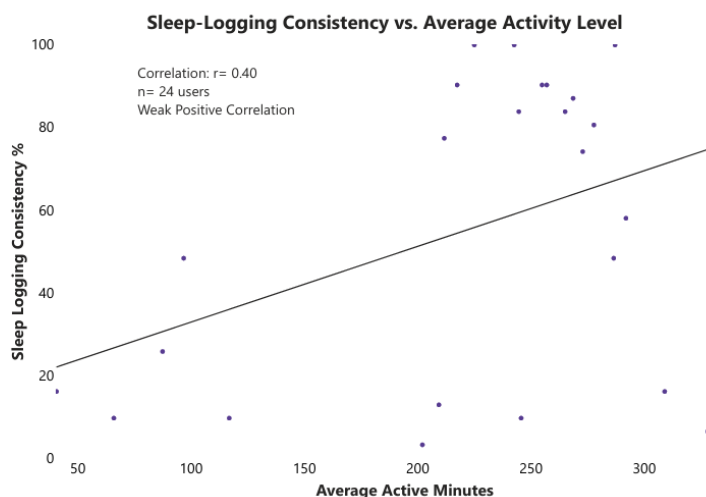
Minute-level data shows 91.5% genuine sleep, 7.5% restless, 1.1% awake — a healthy overall pattern, based on users with sufficient logged data (≥ 5 hours).



Average Minutes Awake in Bed by User ID

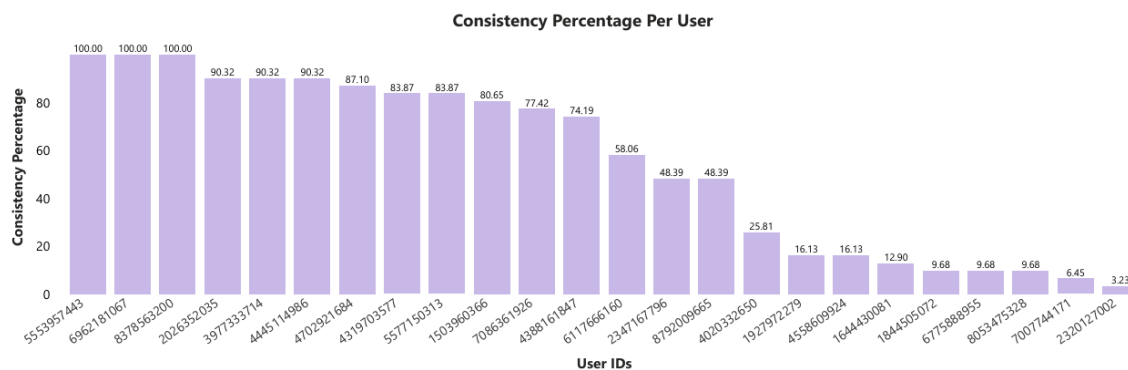
Most of the 24 users show a modest 0–50 minute gap between time in bed and time asleep. Two users stand out with notably larger gaps (309 and 168 minutes), flagged as individual cases of possible sleep difficulty rather than a population-wide trend.

Engagement Pillar



Sleep-Logging Consistency vs. Average Activity Level

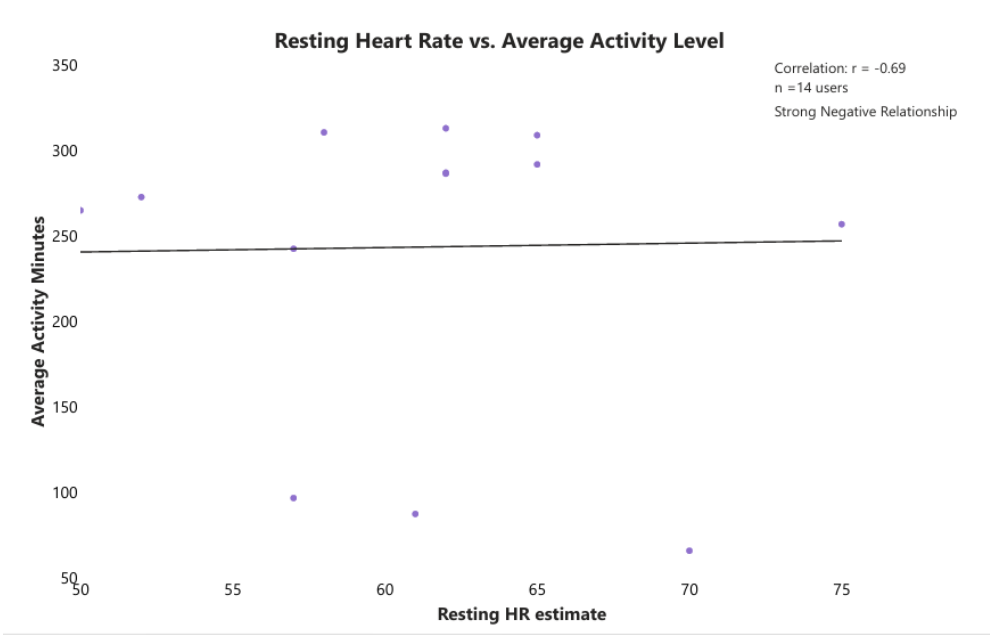
Scatter plot, $r = 0.40$ (weak-to-moderate positive), $n = 24$. Users who log sleep more consistently tend to be somewhat more active, though the relationship isn't strong enough to treat consistency as a reliable predictor on its own.



Consistency Percentage Per User

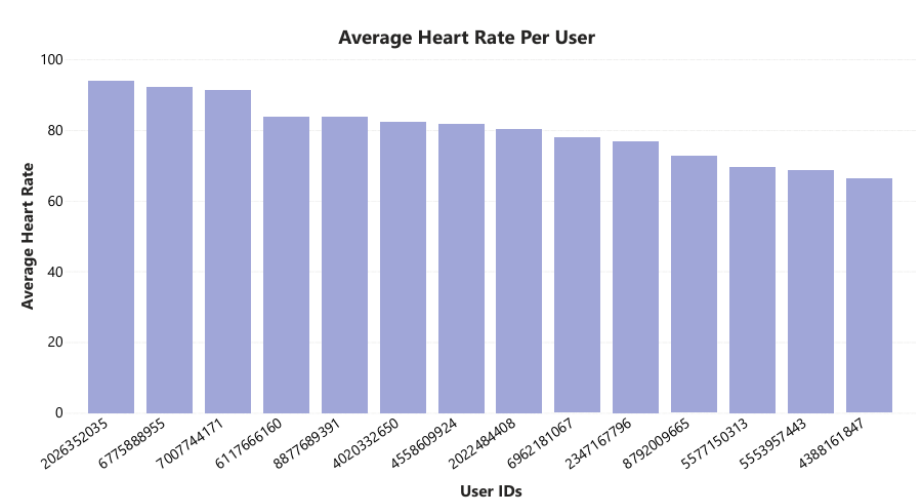
Per-user sleep-logging consistency ranges from 100% down to 3.2%. A meaningful cluster of users log very inconsistently, reinforcing the case for engagement-boosting features (reminders, streaks) rather than assuming tracking habits are self-sustaining.

Heart Rate Deep-Dive



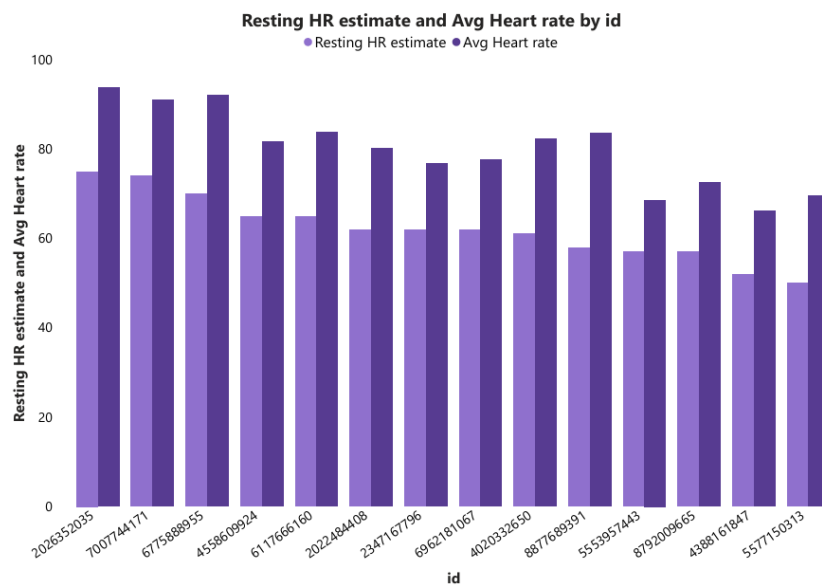
Resting Heart Rate vs. Average Activity Level

Scatter plot, $r = -0.69$, $n = 14$. The dashboard's strongest relationship — more active users have lower resting heart rates, consistent with well-established cardiovascular fitness patterns.



Average Heart Rate Per User

Baseline view of each of the 14 heart-rate-tracking users' overall average heart rate, shown before applying the more robust resting-rate estimate below.



Resting HR Estimate vs. Avg Heart Rate by User ID

Paired comparison confirming resting HR estimates (10th percentile) sit consistently below raw averages for every user — averages are inflated by active periods, while the resting estimate isolates a true calm-state baseline.